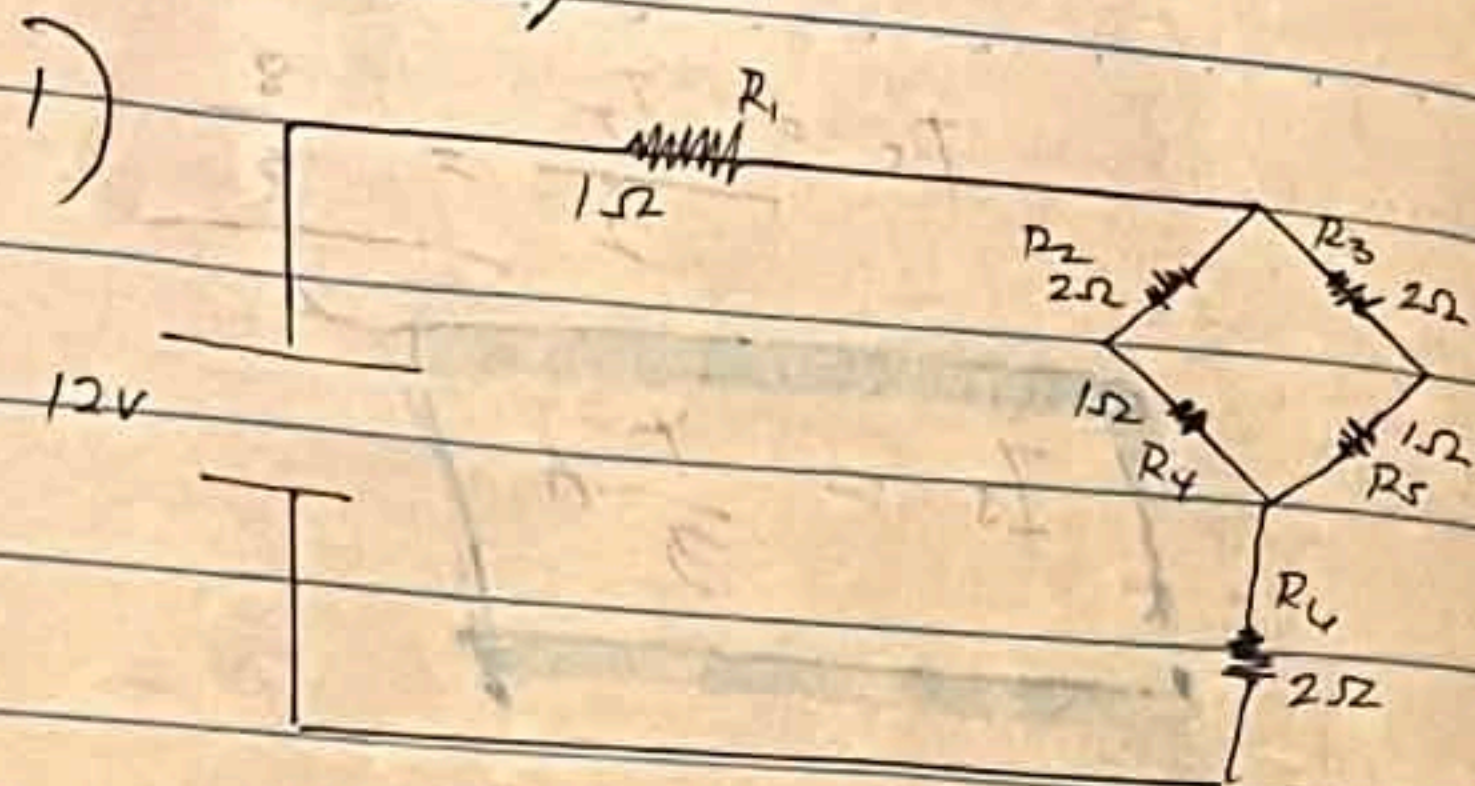


Paula to Esquerres

NO.:

DATE:



$$I_T = \frac{V}{R} = \frac{12V}{\frac{11}{3}\Omega} = \frac{36}{11} A$$

$$V_{2345} = (I_T)(R_{2345})$$

$$= \left(\frac{36}{11} A\right) \left(\frac{2}{3} \Omega\right)$$

$$V_{2345} = \frac{24}{11}$$

Solve for R_T , I_T , V_2 , and I_5

$$R_{24} = R_2 + R_4$$

$$R_{35} = R_3 + R_5$$

$$= 2 + 1$$

$$= 2 + 1$$

$$R_{24} = 3\Omega$$

$$R_{35} = 3\Omega$$

$$I_5 = \frac{V_{2345}}{R_{35}} = \frac{24/11}{3}$$

$$I_5 = \frac{8}{11}$$

$$R_T = R_1 + R_{2345} + R_6$$

$$R_{2345} = \frac{1}{R_{24}} + \frac{1}{R_{35}}$$

$$= \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \Omega$$

$$I_5 = I_2$$

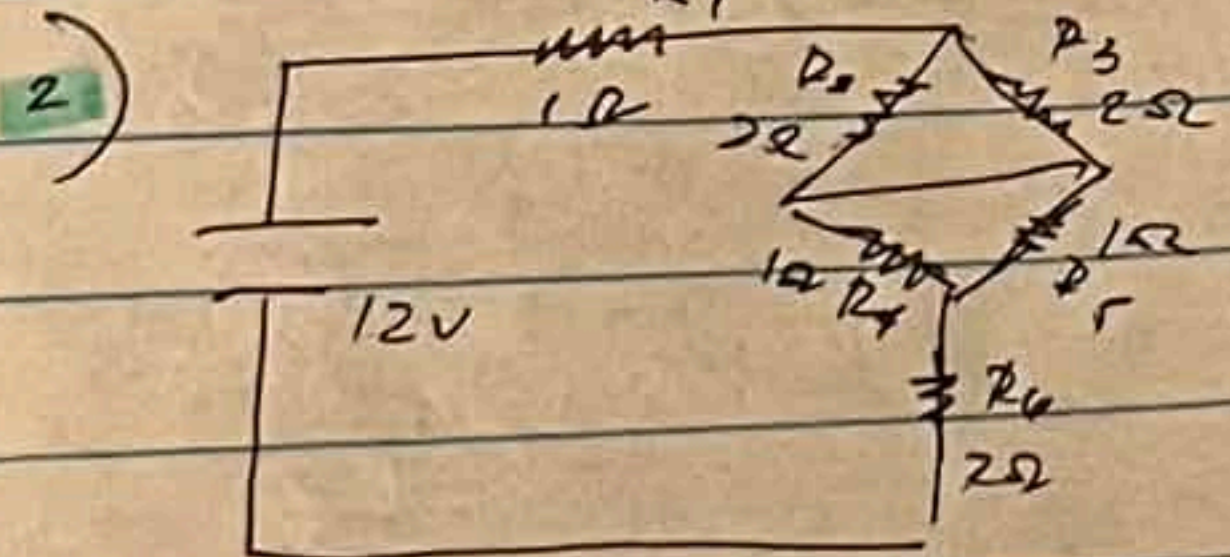
$$V_2 = (I_2)(R_2)$$

$$= \left(\frac{8}{11}\right)(2)$$

$$R_T = 1\Omega + \frac{2}{3}\Omega + 2\Omega$$

$$V_2 = \frac{16}{11}$$

$$R_T = \frac{11}{3} \Omega$$



$$R_{24} = R_2 + R_4$$

$$= 2\Omega + 1\Omega$$

$$R_{35} = R_3 + R_5$$

$$= 2\Omega + 1\Omega$$

$$R_{24} = 3\Omega$$

$$R_{35} = 3\Omega$$

$$R_{2345} = \frac{(R_{24})(R_{35})}{(R_{24}) + (R_{35})} = \frac{9}{6} \Omega$$

$$R_{23} = \frac{R_2}{R_3} = \frac{2}{2} = 1$$

$$R_T = R_1 + R_{2345} + R_6$$

$$= 1 + \frac{9}{6} + 2$$

$$R_{45} = \frac{R_4}{R_5} = \frac{1}{1} = 1$$

$$R_T = \frac{9}{2} \Omega$$

$$I_T = \frac{V}{R_T}$$

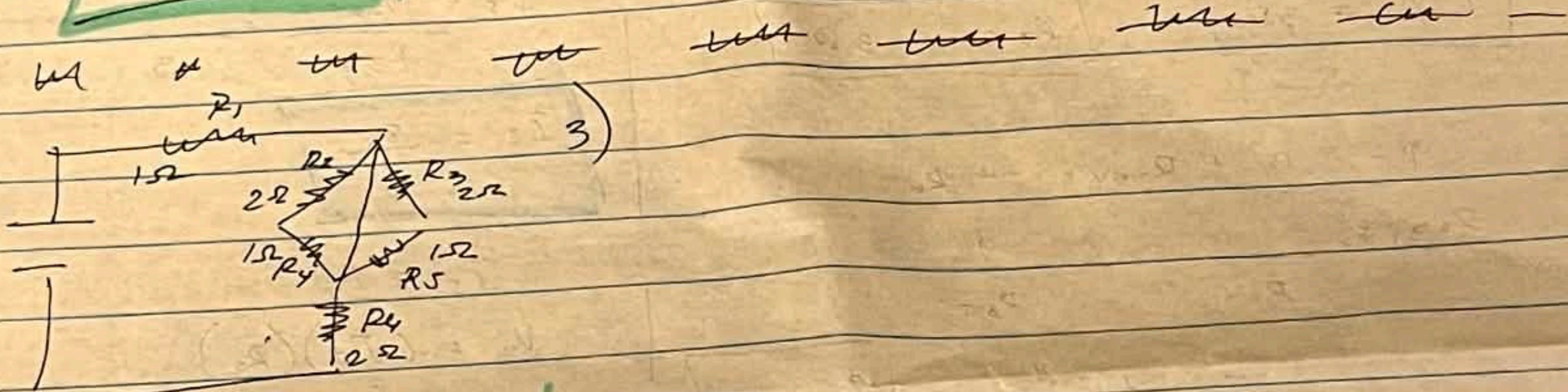
$$= \frac{12V}{\frac{9}{2}} = \frac{24}{9} = \frac{8}{3}$$

$$I_S = \frac{I_T}{2} = \frac{\frac{8}{3}}{2}$$

$$I_S = \frac{4}{3} A$$

$$V_2 = (I_T) \cdot (R_{23}) = \left(\frac{8}{3} A\right) (1.5 \Omega)$$

$$V_2 = \frac{9}{3} V$$



$$R_T = R_1 + R_4$$

$$= 1.5 \Omega + 2 \Omega$$

$$V_2 = 0V$$

$$R_T = 3.5 \Omega$$

$$I_S = 0A$$

$$I_T = \frac{V}{R_T}$$

$$= \frac{12V}{3 \Omega}$$

$$I_T = 4A$$